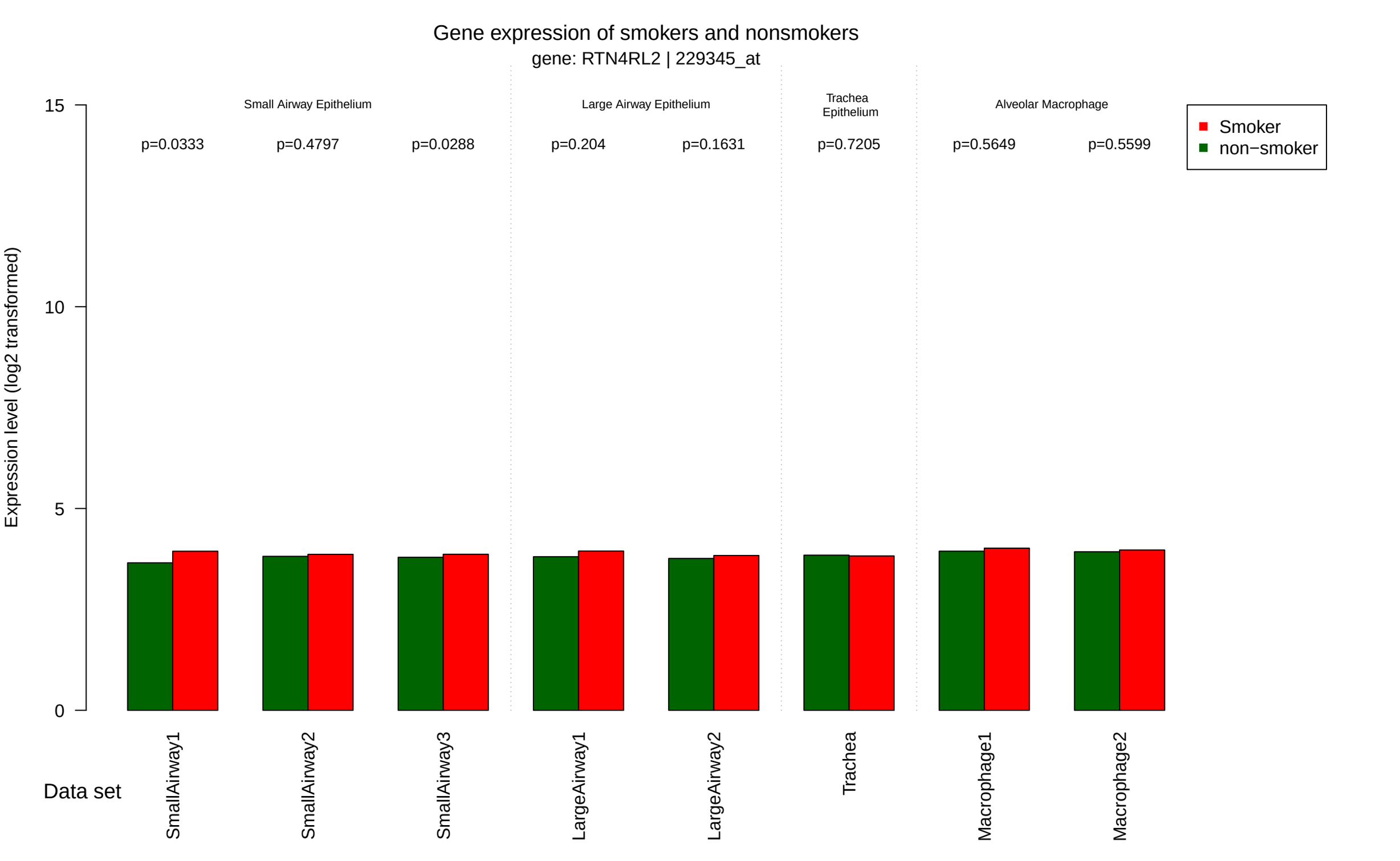
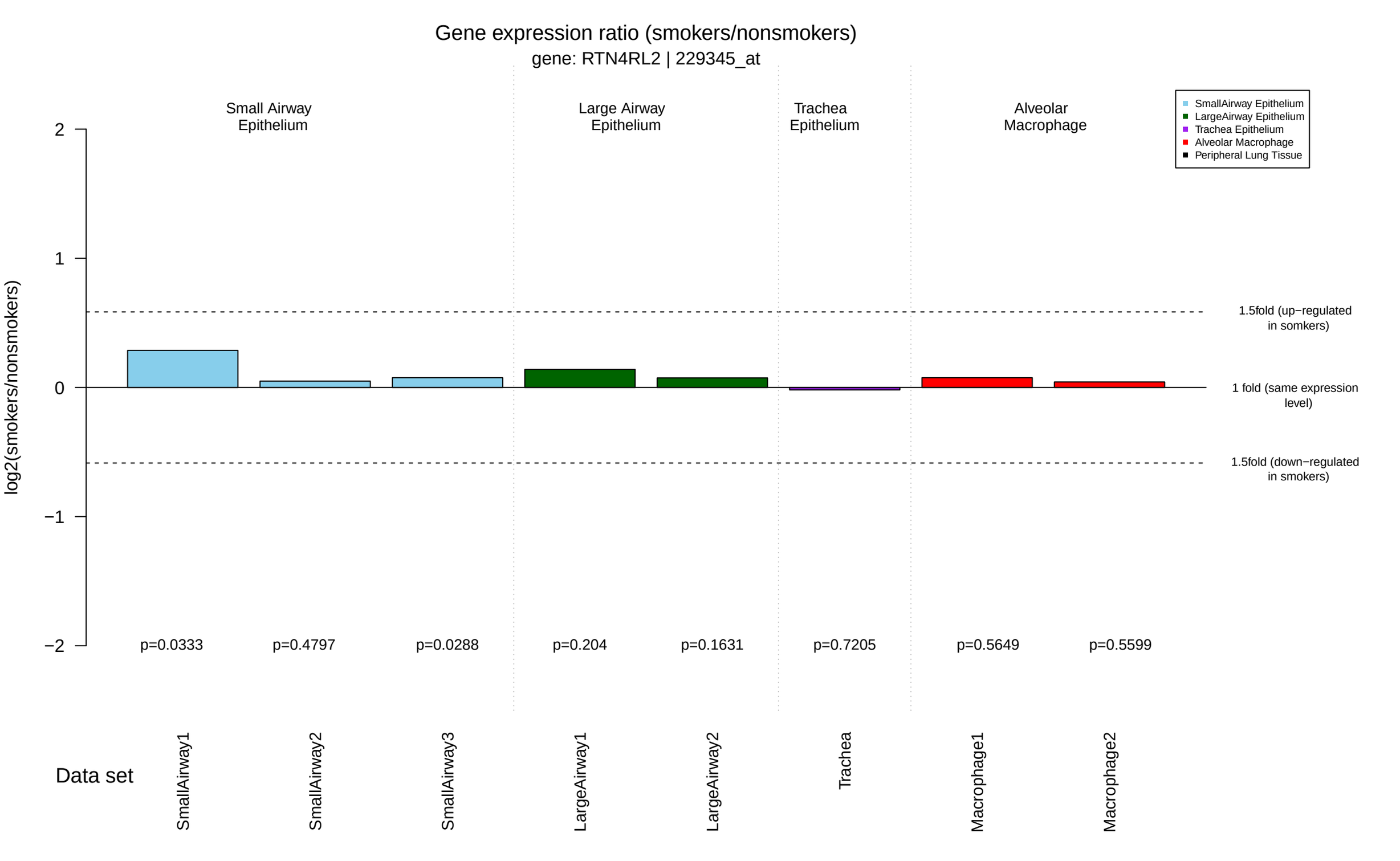
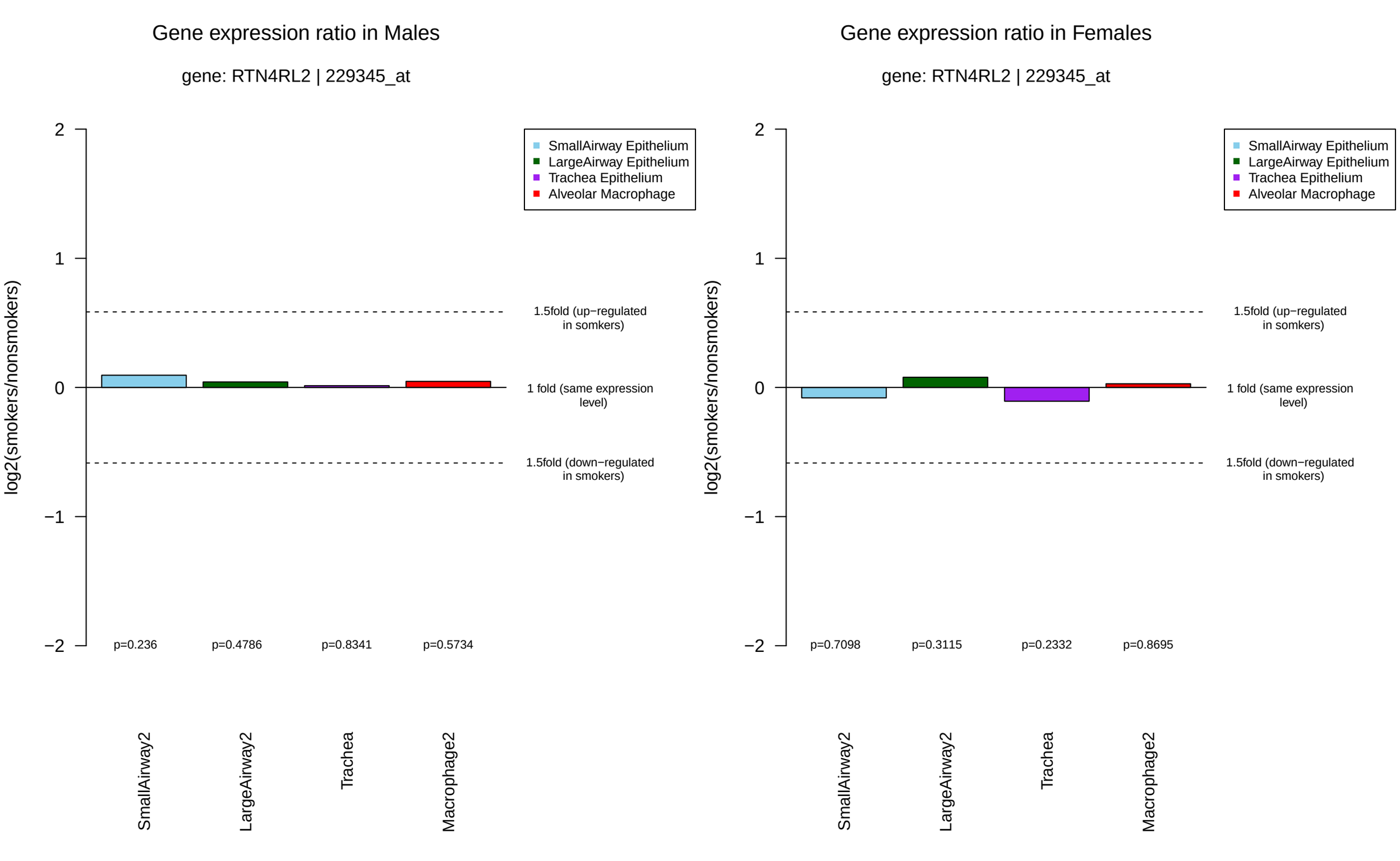
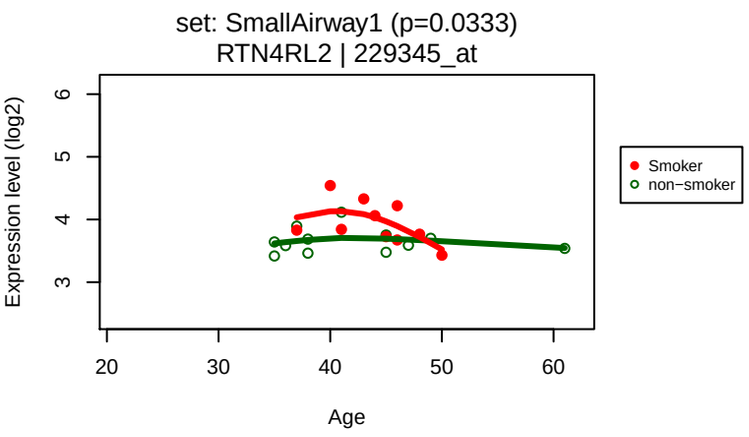
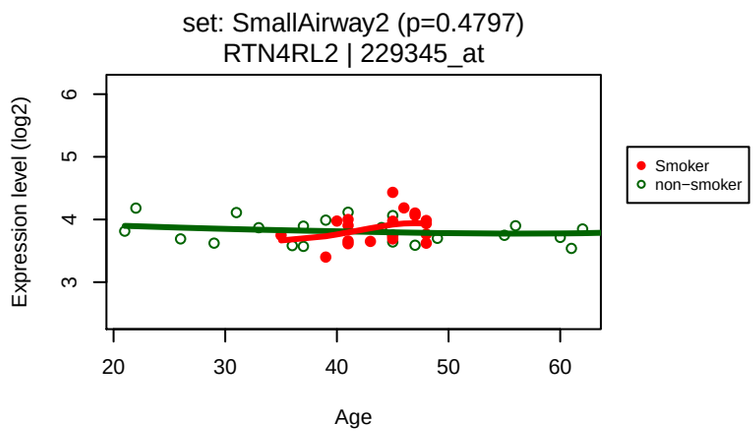
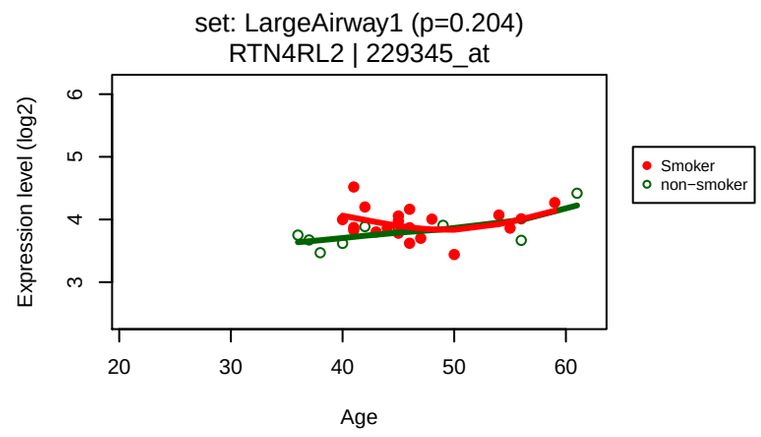

Gene expression of smokers and nonsmokers
gene: RTN4RL2 | 229345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0333
p=0.4797
p=0.0288
p=0.204
p=0.1631
p=0.7205
p=0.5649
p=0.5599
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

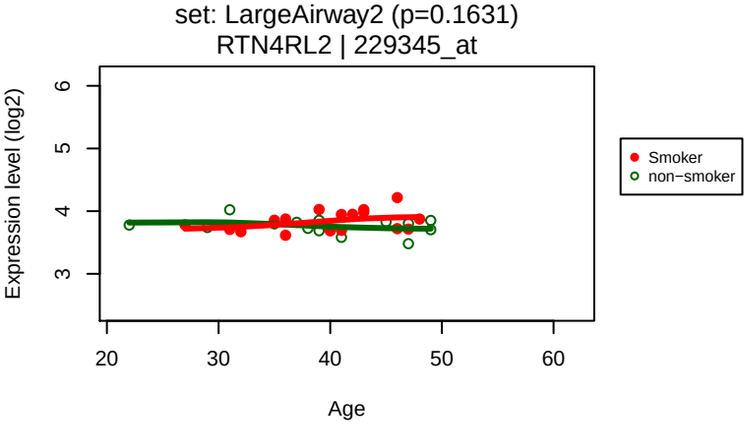

Gene expression ratio (smokers/nonsmokers)
gene: RTN4RL2 | 229345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0333
p=0.4797
p=0.0288
p=0.204
p=0.1631
p=0.7205
p=0.5649
p=0.5599
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

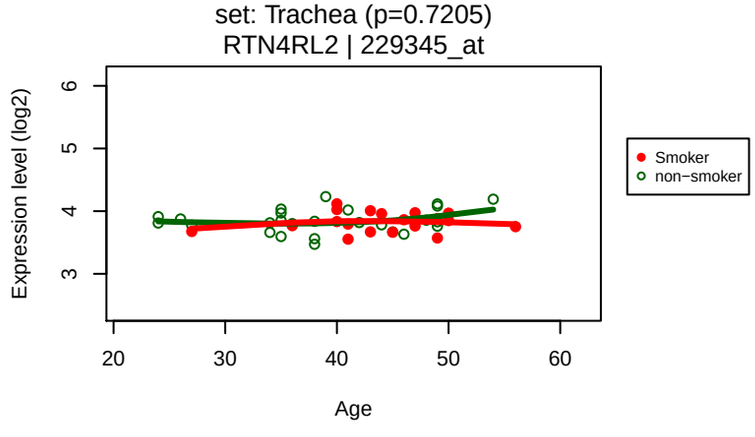

Gene expression ratio in Males
gene: RTN4RL2 | 229345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.236
p=0.4786
p=0.8341
p=0.5734
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTN4RL2 | 229345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7098
p=0.3115
p=0.2332
p=0.8695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

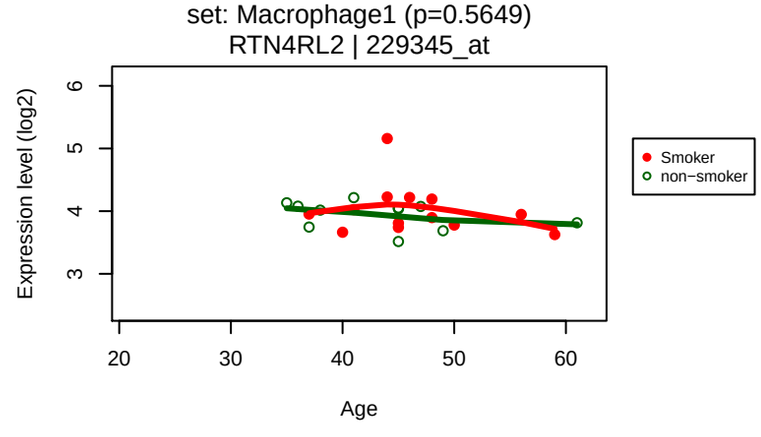

set: SmallAirway1 (p=0.0333)
RTN4RL2 | 229345_at
Expression level (log2)
Age
Smoker
non−smoker

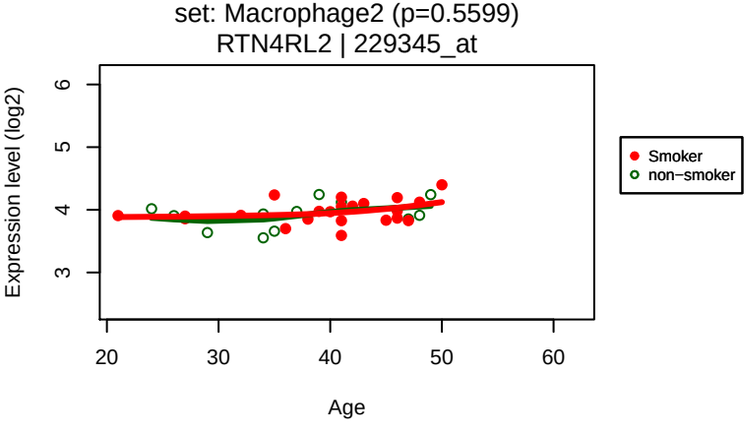

set: SmallAirway2 (p=0.4797)
RTN4RL2 | 229345_at
Expression level (log2)
Age
Smoker
non−smoker

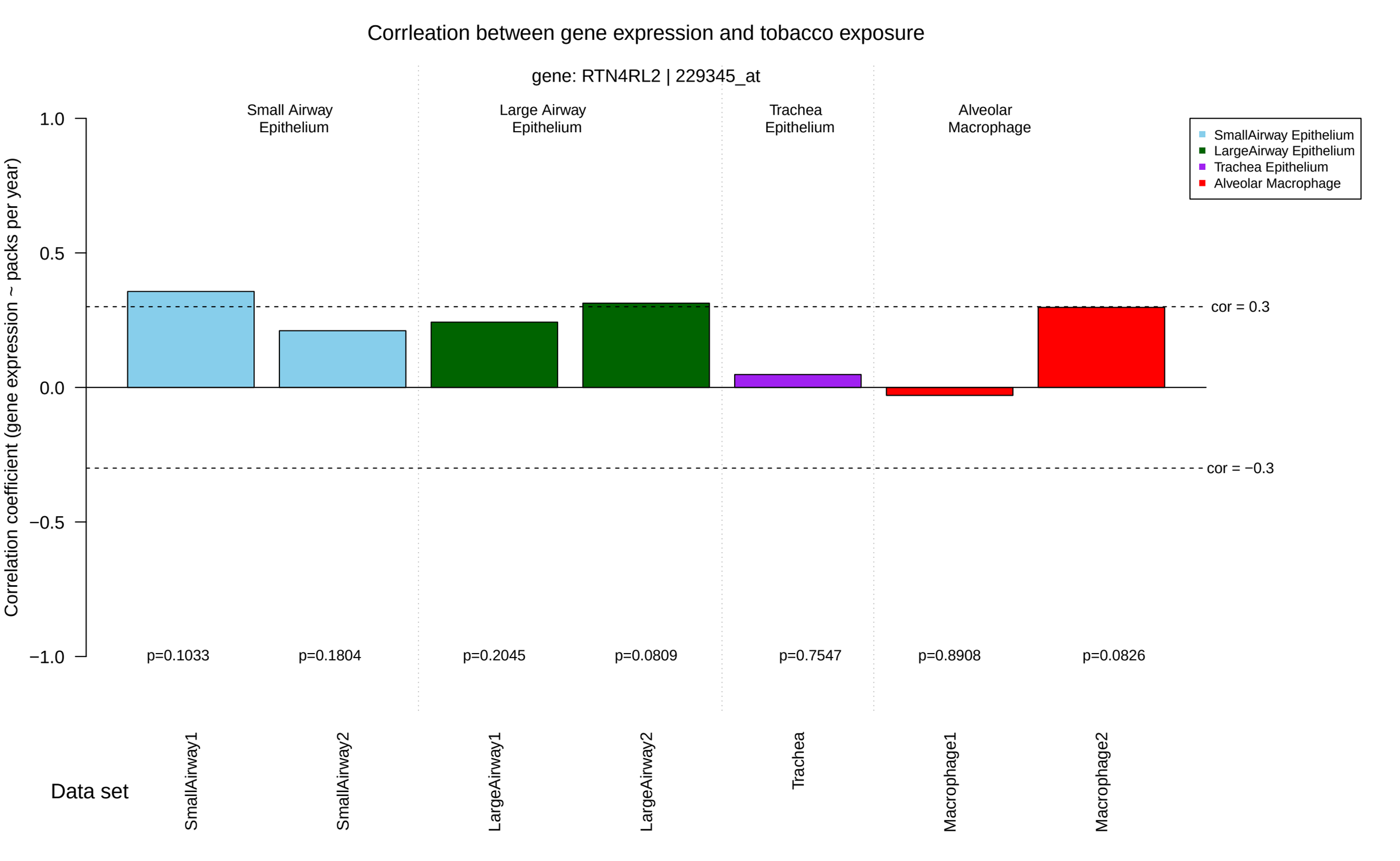

set: LargeAirway1 (p=0.204)
RTN4RL2 | 229345_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1631)
RTN4RL2 | 229345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7205)
RTN4RL2 | 229345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5649)
RTN4RL2 | 229345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5599)
RTN4RL2 | 229345_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTN4RL2 | 229345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1033
p=0.1804
p=0.2045
p=0.0809
p=0.7547
p=0.8908
p=0.0826
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2